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Myelin: A possible proton capacitor for energy storage during sleep and energy supply during wakefulness

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ABSTRACT

There are several physiological reasons why biological organisms sleep. One key one concerns brain metabolism. In our article we discuss the role of metabolism in myelin, based on the recent discovery that myelin contains mitochondrial components that enable the production of adenosine triphosphate (ATP) via oxidative phosphorylation (OXPHOS). These mitochondrial components in myelin probably originate from vesiculation of the mitochondrial membranes in form from mitochondrial derived vesicles (MDVs). We hypothesize that myelin acts as a proton capacitor, accumulating energy in the form of protons during sleep and converting it to ATP via OXPHOS during wakefulness. Empirical evidence supporting our hypothesis is discussed, including data on myelin metabolic activity, MDVs, and allometric scaling between white matter volume and sleep duration in mammals.

1. Introduction

Sleep is widely acknowledged to be one of the most enigmatic phenomena in biology (Frank, 2006). While several physiological reasons for sleeping have been discovered (including tissue repair, neuronal information consolidation, brain plasticity and metabolic by-product removal from the brain) (Brodt et al., 2023; Frank and Heller, 2019; Hauglund et al., 2020; Krueger et al., 2016; Lewis, 2021), the main reasons for sleep remain largely elusive. One of the most widely accepted hypotheses is that sleep serves to restore the brain's energy reserves (Benington and Heller, 1995; Scharf et al., 2008). The hypothesis was initially linked to brain glycogen; however, this is now being questioned given that glycogen is only present in minimal quantities in the brain. Nevertheless, the hypothesis that sleep serves to accumulate energy remains widely accepted (Berger and Phillips, 1995; Latifi et al., 2018; Schmidt, 2014; Schmidt et al., 2017). In this context, the term "energy" is associated with mitochondria, and the significance of mitochondria for sleep is increasingly discussed (Hartmann and Kempf, 2023; Richardson and Mailloux, 2023; Sarnataro et al., 2024). Concurrently, studies have documented the identification of specific myelin genes whose expression is modified during sleep (Bellesi et al., 2013). It is noteworthy that myelin undergoes alterations during sleep (Bellesi et al., 2018; de Vivo and Bellesi, 2019; Toschi et al., 2021), leading de Vivo and Bellesi to conclude that "one of the functions of sleep may be indeed to maintain healthy white matter" (de Vivo and Bellesi, 2019). It can therefore be proposed that there is a link between sleep, mitochondria and myelin. Furthermore, variations in adenosine triphosphate (ATP) levels in cortical excitatory neurons during sleep have been reported (Natsubori et al., 2020), which is in line with previous observations (Dworak et al., 2010; Wong-Riley, 2011).

This article discusses the role of myelin and cerebral metabolism in sleep, incorporating the recent discovery that myelin incorporates mitochondrial components that traverse the processes of the oligodendrocyte, becoming embedded in the myelin during its compaction providing ATP by oxidative phosphorylation (OXPHOS) (Morelli et al. 2011b, 2011c). The precise process has yet to be established in detail, but we suggest that the mitochondrial components originate from vesiculation of the mitochondrial membranes, i.e., mitochondrial

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derived vesicles (MDVs). These are widespread among eukaryotes and are also present in the brain (Hazan Ben-Menachem et al., 2024; Kim et al., 2024). Recently we demonstrated that MDVs contain a functional FoF1-ATP synthase and are capable of independent ATP synthesis (Hazan Ben-Menachem et al., 2023).

Based on these premises, we have formulated a hypothesis that myelin functions as an energy accumulator in the form of protons stored in laminar acid/base buffers. During sleep, these protons would be charged, and during wakefulness, they would be discharged. This hypothesis has previously been proposed in broad terms (Morelli et al., 2011a; Morelli, 2019) and has recently been employed to elucidate the actions of white widow toxins (Gasanoff, 2024) and the cobra snake (Gasanoff and Dagda, 2024). In our article we present an approximate calculation of the number of protons that can accumulate during sleep in the myelin proteolipid protein (PLP) and myelin basic protein (MBP), which are present in high concentrations in the myelin sheath. The results of our calculations indicate that the human brain needs ~8 h of sleep to be recharged by the oxidative metabolism that is active in myelin.

2. Myelin and the embedded proteins PLP and MBP

Myelin is formed by specific glial cell types (oligodendrocytes in the central nervous system [CNS] and Schwann cells in the peripheral nervous system [PNS]) that generate a multilayer sheath around nerve cell axons (Kister and Kister, 2022) (Fig. 1(a)).

Myelin-bound water interacts with protons in the myelin macromolecules leading to specific MRI fast relaxation signal properties. The myelin content of the human brain can be imaged non-invasively based on this principle by for example quantifying the magnetization exchange from the water within the myelin sheath to the water outside of it (Ouellette et al., 2020) (Fig. 1(b)).

The hypothesis presented in this article is based on the well-founded assumption that the myelin sheath may have a proton capacitor

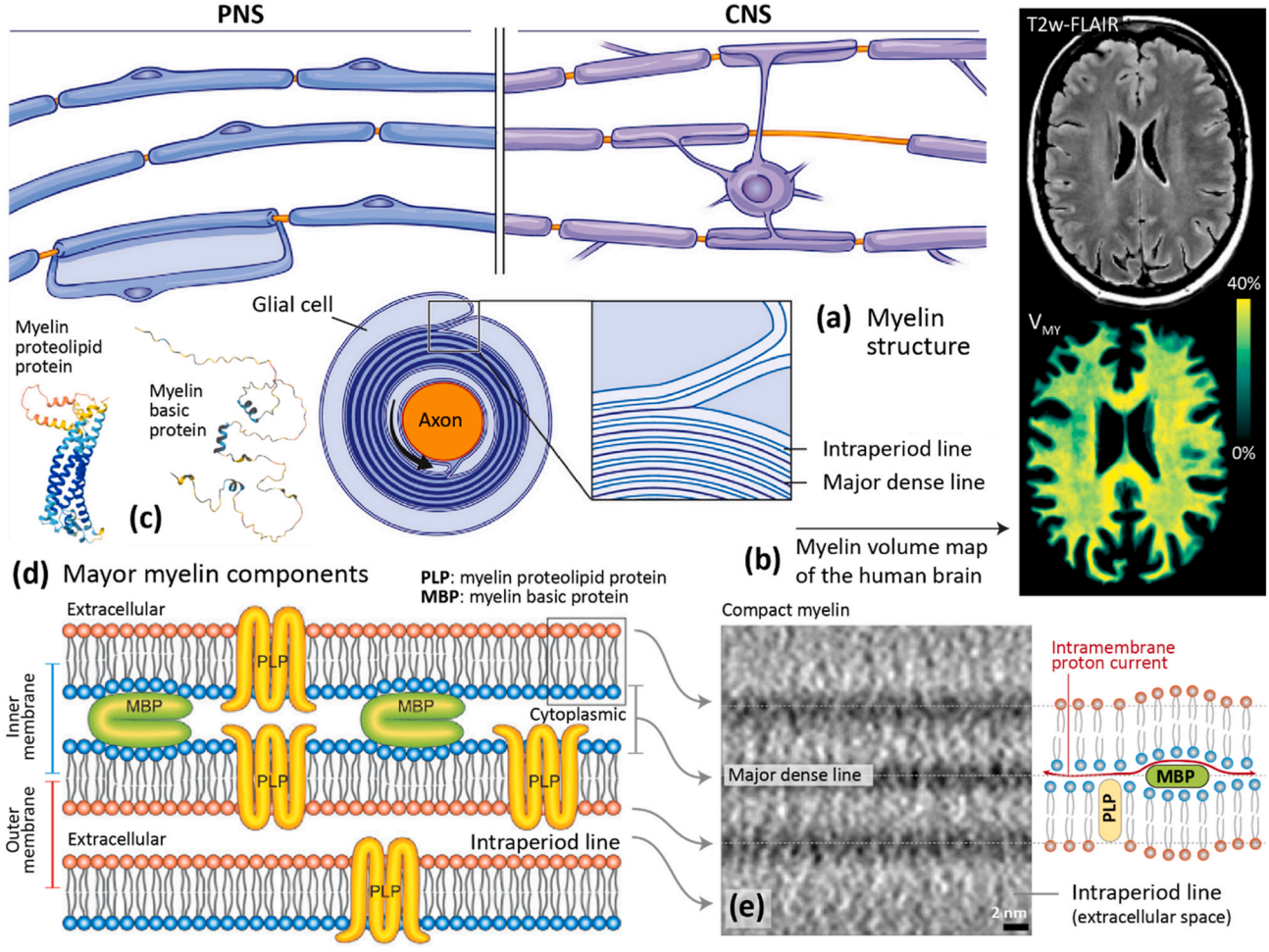


Fig. 1. Structure and composition of myelin. (a) Myelination in the peripheral (PNS) and central nervous system (CNS) with individual myelin sheaths (blue) formed by Schwann cells around axons (orange) in the PNS, and multiple myelin sheaths (purple) formed by oligodendrocytes in the CNS. The cross-section of a myelin sheath visualizes the origin of the intraperiod line and major dense line of myelin. Reprinted and modified from Salzer and Zalc (Salzer and Zalc, 2016), with permission from the publisher. **(b)** Myelin partial volume (V_{MY}) image of the human brain along a T2-weighted fluid-attenuated inversion recovery (T2w-FLAIR) image. Both obtained with magnetic resonance imaging (MRI) at 3.0 T. Example images of a 56-year-old healthy female. Reprinted and modified from Ouellette et al. (Ouellette et al., 2020), with permission from the publisher. **(c)** Computed structure model of myelin basic protein (MBP) (PDB protein data bank, PDB ID: AF_AFP02687F1) and myelin proteolipid protein (PLP) (PDB protein data bank, PDB ID: AF_AFP60201F1) (Berman et al., 2000). **(d)** Structure of the myelin sheath with the main myelin components. The apposition of the inner surfaces of the myelin bilayer forms the major dense line, while the intraperiod line is formed by the apposition of the outer surfaces of the myelin bilayer. Reprinted and modified from Aggarwal et al. (Aggarwal et al., 2011), with permission from the publisher. **(e)** Ultrastructure of human brain myelin obtained by cryo-electron tomography. The major dense line and the extracellular-facing polar head group densities are clearly visible. Reprinted and modified from Creekmore et al. (Creekmore et al., 2024), with permission from the publisher. A schematic of the myelin architecture with two bilayers and the embedded MBP and PLP proteins is shown on the right. The amino acids serine and threonine of the PLP are excellent proton conductors and enable a proton current in the major dense line. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

function, as indicated by a number of findings, including the detailed structure of PLP and MBP and how they are incorporated into the myelin layers. In particular, we refer to the findings of Petri Kursula's group and their detailed description of the placement of the embedded MBP in the major dense line (Raasakka et al., 2017) and of the PLP that connects the aqueous phase of the major dense line with the aqueous phase of the intraperiod line (Greer and Lees, 2002; Ruskamo et al., 2022). Further detailed investigations about the MBP embedded in the major dense line were performed by Omlin et al., (1982) and Hu et al., (2004). The major dense lines (width: about 2–3 nm) and the intraperiod lines (width: about 4 nm) are specific morphological features of the myelin sheath (Kister and Kister, 2022). The MBP-PLP association fulfils the functions of both a proton capacitor (primarily the responsibility of MBP) and a proton conductor (performed exclusively by PLP). Given its fundamental composition, MBP appears capable of absorbing protons resulting from the oxidation of nicotinamide-adenine dinucleotide (NADH), which is carried out by respiratory chain complex I (NADH biquinone oxidoreductase) during the energy recharging phase, or sleep phase.

The MBP can be considered a buffer due to its horizontal U-shaped structure, in which the amino acids lysine and arginine are situated on the two arms of the structure (Fig. 1(c)). MBP synthesis occurs in oligodendrocyte processes (Gould and Brady, 2023) and the MBP protein remains embedded between the two plasma membranes of the oligodendrocyte processes, which is forming the primary dense line of compact myelin. Fig. 1(e), depicts a schematic representation of compact myelin, wherein three major dense lines are delineated, separated by the intraperiod zone. Fig. 1(d) depicts the ultrastructure of myelin with the inner and outer myelin layer as well as the embedded MBP and PLP.

During wakefulness, the protons leave laminar buffers consisting of MBP, the major dense line basic domain of PLP and nitrogenous compounds, such as choline, which esterify the phosphate residue of phospholipids. This process occurs in the Fo rotor of the FoF1-ATP synthase, resulting in the subsequent synthesis of adenosine triphosphate (ATP). The protons exiting the Fo rotor flow in the intraperiod line thanks to intramembrane proton conductor domains of the PLP, which contain a high percentage of serine and threonine amino acids. These amino acids are efficient formers of proton wires (Kitazoe et al., 2008; Nagle and Morowitz, 1978; Panfoli et al., 2017) through which the protons exiting the Fo rotor can reach the basic domain of the PLP immersed in the intraperiod line (see section 4 for details). We hypothesize that the accumulation of protons in the MBP corresponds to the sleep phase, while the release of protons and the subsequent synthesis of ATP correspond to the waking phase.

3. Functional coupling of extra-mitochondrial OXPHOS and myelin proteins in the myelin sheath

The capacity of myelin to synthesize ATP through the conventional OXPHOS, despite the absence of mitochondria (Rinholm et al., 2016), is substantiated by numerous experimental findings (Morelli et al. 2011b; Ravera et al. 2009, 2013, 2016a, 2016b, 2020).

Myelin is therefore good example of a site of extra-mitochondrial OXPHOS. Even in the peripheral nervous system, despite the minimal presence of mitochondria, consistent ATP synthesis occurs in myelin (Hong et al., 2024). In our opinion, the respiratory complexes generate proton currents with proton movements without ever leaving the proteolipid phase, which is a phase very different from the cytosol or aqueous phase (Mulkidjanian and Cherepanov 2006). In other words, the protons most likely pass directly from dissociable groups to other dissociable groups, without moving freely, as their movement would otherwise result in the formation of areas of high acidity and enormous destructive power for any biological structure. The claim that protons must never be released is supported by the evidence that biological membranes are impermeable to all substances except protons (Bozdaganyan et al., 2019; Grzesiek and Dencher, 1986; Tepper and Voth, 2005). Consequently, if protons were released, they would immediately leave the membrane.

Fig. 2 presents two crucial concepts for our further analysis, which is inspired by a myelin model in which OXPHOS is divided into two phases, as previously described (Gasanoff and Dagda, 2024; Morelli, 2019): (i) an initial *redox coupling* phase in which proton movement is driven by the electron transport chain (ETC), which involves the oxidation of reducing equivalents (typically NADH or dihydroflavine-adenine dinucleotide [$FADH_2$]) coupled reduction of oxygen (O_2), and protonation of the MBP (see Fig. 2(a)), and (ii) a second *proton/ATP synthase coupling* phase, in which proton movement from the MBP buffers is coupled to ATP synthesis by FoF1-ATP synthase (see Fig. 2(b)). Interestingly, this aligns with what was observed in MDV, namely that these can generate ATP without external Krebs cycle substrates and that the driving force is solely the intrinsic membrane potential (Hazan Ben-Menachem et al., 2023).

We postulate that the redox coupling is active during sleep, while the proton/ATP synthase coupling characterizes the waking state.

Diurnal variation of mitochondrial function is known in plants, which makes it conceivable that this phenomenon could also occur in animal cells (Lee et al., 2010).

The experimental data obtained in our laboratory corroborate this hypothesis (Ravera et al., 2021). The myelinated vesicles retain an energy charge obtained during exposure to NADH, which is subsequently used for ATP synthesis. This is achieved by subjecting the vesicles, which have been removed from their original NADH exposure, to adenosine diphosphate (ADP) and inorganic phosphate (P_i) in a subsequent phase. According to our measurements, the longer the previous exposure time to NADH, the greater the amount of ATP synthesized. It is therefore plausible to assume that in the first phase of the experiment (see Fig. 2 (a)) the redox coupling was active (corresponding to the sleep phase of "recharging" the myelin proton capacitor), whereas in the second phase of the experiment (see Fig. 2(b)) the proton/ATP synthase coupling was active (corresponding to the wake phase).

The fact that membranes in general can store energy by accumulating protons has recently been considered (Gasanoff, 2024; Gasanoff and Dagda, 2024; Nesterov et al., 2022), which supports our considerations.

4. A possible role of PLP as a proton conductor involved in the formation of a proton current circuit

The most abundant myelin protein in CNS myelin is the fatty acid-esterified integral membrane protein PLP (about 40 %) (Knapp, 1996) with a half-life of ~90 days (Greer and Lees, 2002). PLP is mainly expressed in CNS myelin and a reduced expression is observed in PNS myelin (Greer and Lees, 2002). PLP is expressed in the CNS also in minor isoform (DM-20) which lacks 35 residues in the intracellular loop (see residues 116–150 which are indicated in yellow in Fig. 3(a)) and has an abundance of ~50 % relative to PLP mRNA in the rat brain (Nave et al., 1987). The function of PLP is not fully understood, although severe pathologies due to PLP mutations are known (Khalaf et al., 2022). Despite PLP and MBP, hundreds of other proteins are present in myelin in lower concentrations; 480 non-redundant proteins were identified in the myelin-enriched fraction of human brain samples (Martins-de-Souza et al., 2021).

Interestingly, PLP has been shown to bind specifically to MBP and not to other CNS proteins (Edwards et al., 1989) which is consistent with the cooperative mechanism between the two proteins as proposed by our hypothesis. In addition, PLP has been shown to act as a proton conductor (Lin and Lees, 1982). Given this evidence, we propose that PLP, in conjunction with MBP, contributes to the proton conduction function proposed for myelin, with the addition that PLP favors proton movement because there is a region in PLP that contains optimal amino acids for proton conduction, i.e., serine and threonine (Kitazoe et al., 2008). These serine and threonine residues (circled in blue in Fig. 3(a))

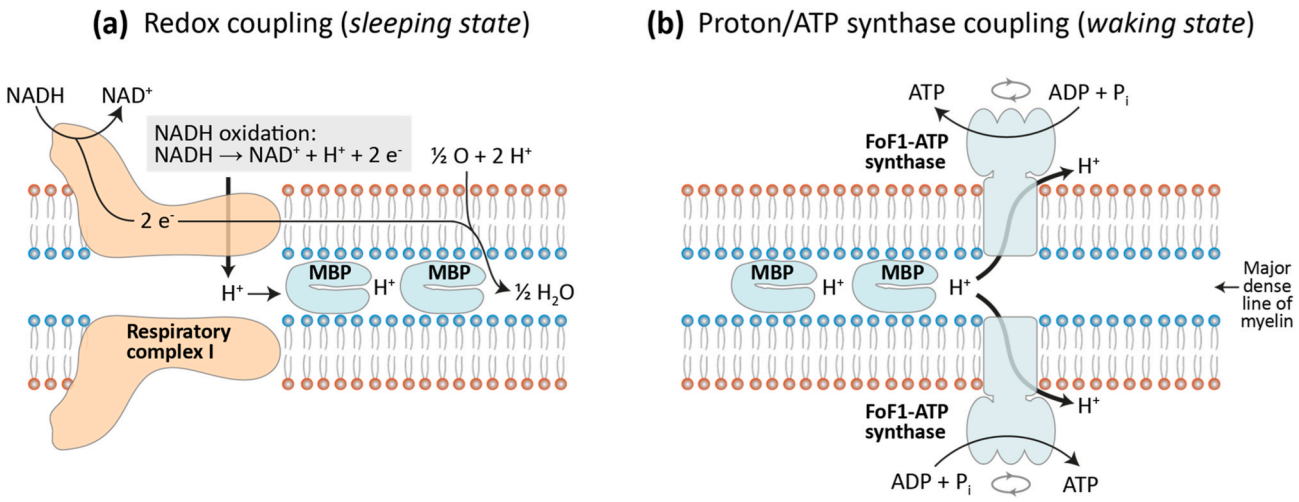


Fig. 2. Two phases of extra-mitochondrial OXPHOS activity in myelin depending on the state of the organism (waking/sleeping). (a) Redox coupling phase. The respiratory complex I couples the oxidation of NADH to the movement of protons (H^+) which "charge" the MBPs by protonating them. The oxidation of NADH generates NAD^+ and a H^+ (which is transported into the membrane to protonate the MBP and PLP) plus 2 electrons (e^-) which support the electron transport chain until the reduction of oxygen (O_2). The major dense line basic domain of the PLP would also be protonated but for simplicity it is not included in the schematic. The process depicted in (a) corresponds to the sleep phase, i.e., the "recharging" of myelin without the production of ATP. For illustrative purposes, only respiratory complex I is shown and the other components of the mitochondrial respiratory chain (i.e. succinate-CoQ reductase, cytochrome bc_1 complex, cytochrome *c* oxidase) are not visualized. **(b)** Proton/ATP synthase coupling which involves the "discharge" of the MBP (and PLP) with the conveyance of a H^+ current to the FoF1-ATP synthase which drives the rotation of the Fo rotor with the consequent synthesis of ATP. This corresponds to the waking phase in which MBP is "discharged" and the generated H^+ current enables the synthesis of ATP. It should be noted that the exact positioning of the MBP and the FoF1-ATP synthase is what makes these two processes possible in the first place: the protonated arms of the MBP are symmetrically aligned to the membranes that for the major dense line, and the F1 subunit is placed in the two intraperiod lines on the side of the major dense line where ATP synthesis occurs in the aqueous phase.

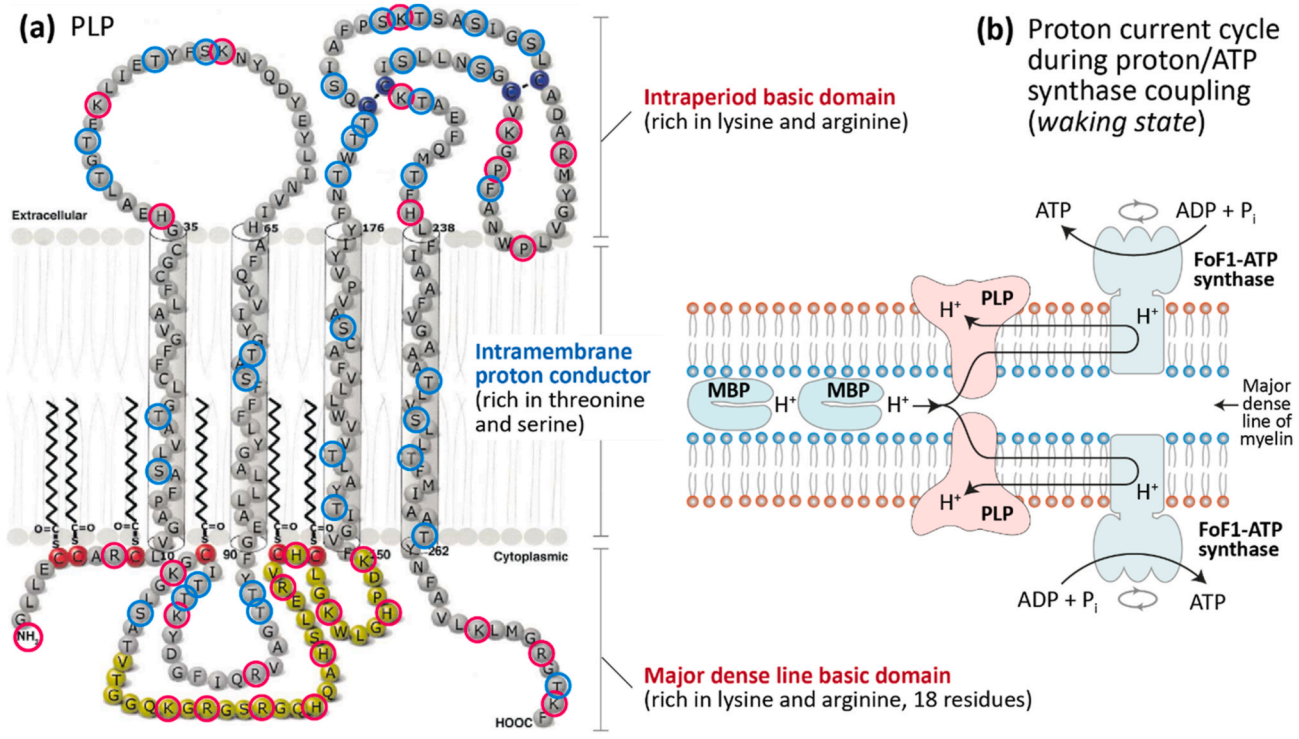


Fig. 3. The role of the PLP in the proton current circuit of the myelin proteins and the FoF1-ATP synthase. (a) Structure of PLP and its position in myelin in the lipid bilayer connecting the major dense line and the intraperiod line. Four transmembrane α-helices of PLP pass through the membrane. The basic amino acids lysine and arginine are highlighted in red and are present in high percentages only in the two domains located in the major dense line and in the intraperiod line. The amino acid residues of serine and threonine (which are excellent proton conductors) are indicated in blue and are located in the transmembrane tracts. The sections of PLP that are absent in the minor PLP isoform DM-20 (residues 116–150) are indicated in yellow. Reprinted and modified from Greer and Lees (Greer and Lees, 2002), with permission from the publisher. **(b)** Schematic representation of the postulated proton flow during the proton/ATP synthase coupling phase (waking state) between the myelin proteins (MBP and PLP) and the FoF1-ATP synthase, resulting in the propulsion of Fo rotor. It is postulated that protons flow from the MBP to the intraperiod basic domain of the PLP and the further to the FoF1-ATP synthase. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

form the intramembrane proton-conducting domain (Fig. 1(c)) which we hypothesize to exist in myelin. Furthermore, PLP has a domain immersed in the cytosolic phase of the major dense line (see intraperiod basic domain, Fig. 1(d)), rich in the basic amino acids lysine and arginine (circled in red in Fig. 3(a)) and a major dense line basic domain immersed in the aqueous phase of the intraperiod line (see Fig. 1(d) and **3(a)**).

Thus, PLP has two basic domains, one in the major dense line and one in the intraperiod line, connected by the proton-conducting intramembrane domain. It is important to note that both the proton-conducting amino acids (serine and threonine) and the basic amino acids are distributed in very different domains of PLP, supporting the functions proposed here. Since the intraperiod basic domain is rich in both basic amino acids and threonine and serine, we propose that this domain functions both of a proton conductor and a proton capacitor (buffer).

In light of this distribution of amino acids, we furthermore propose that a closed proton circuit is formed at the major dense line (see Fig. 3 (b)): the respiratory complex, by oxidizing NADH, accumulates protons during redox coupling (i.e. during sleep) in the MBP and also in the area immersed in the major dense line of the PLP and the nitrogenous compounds (i.e., choline moiety) of the phospholipid heads. During wakefulness, the protons from MBP and PLP are transferred to the rotor of ATP synthase; ATP synthesis takes place, and the protons are transported along the membrane from the threonine- and serine-rich PLP region to accumulate in the PLP region immersed in the aqueous phase of the intraperiod line.

The existence of a PLP-Fo rotor interaction is supported by the fact that the isolation of PLP led to the co-purification of Fo (Benjamins et al., 1994), so much so that a contribution of PLP to the synthesis of ATP by FoF1-ATP synthase has been hypothesized (Lees et al., 1979).

The topic of myelin PLP and mitochondrial PLP is closely related, as myelin PLP deficiency not only causes hereditary spastic paraplegia, but also leads to mitochondrial dysfunction (Yin et al., 2016).

In Fig. 2, for simplicity, we show only respiratory complex I, but in reality it is the respiratory chain complexes I-II-III-IV that constitute the ETC (Hirst, 2018; Lenaz and Genova, 2009; Letts et al., 2016). However, complex I is the only one that interacts with NADH and for this purpose it also has a protuberance directed towards the aqueous phase of the intraperiod line (Morelli et al., 2020) and we can speculate that it can extract protons from the intraperiod basic domain of the PLP to transfer them to the MBP, thus resuming the proton circuit.

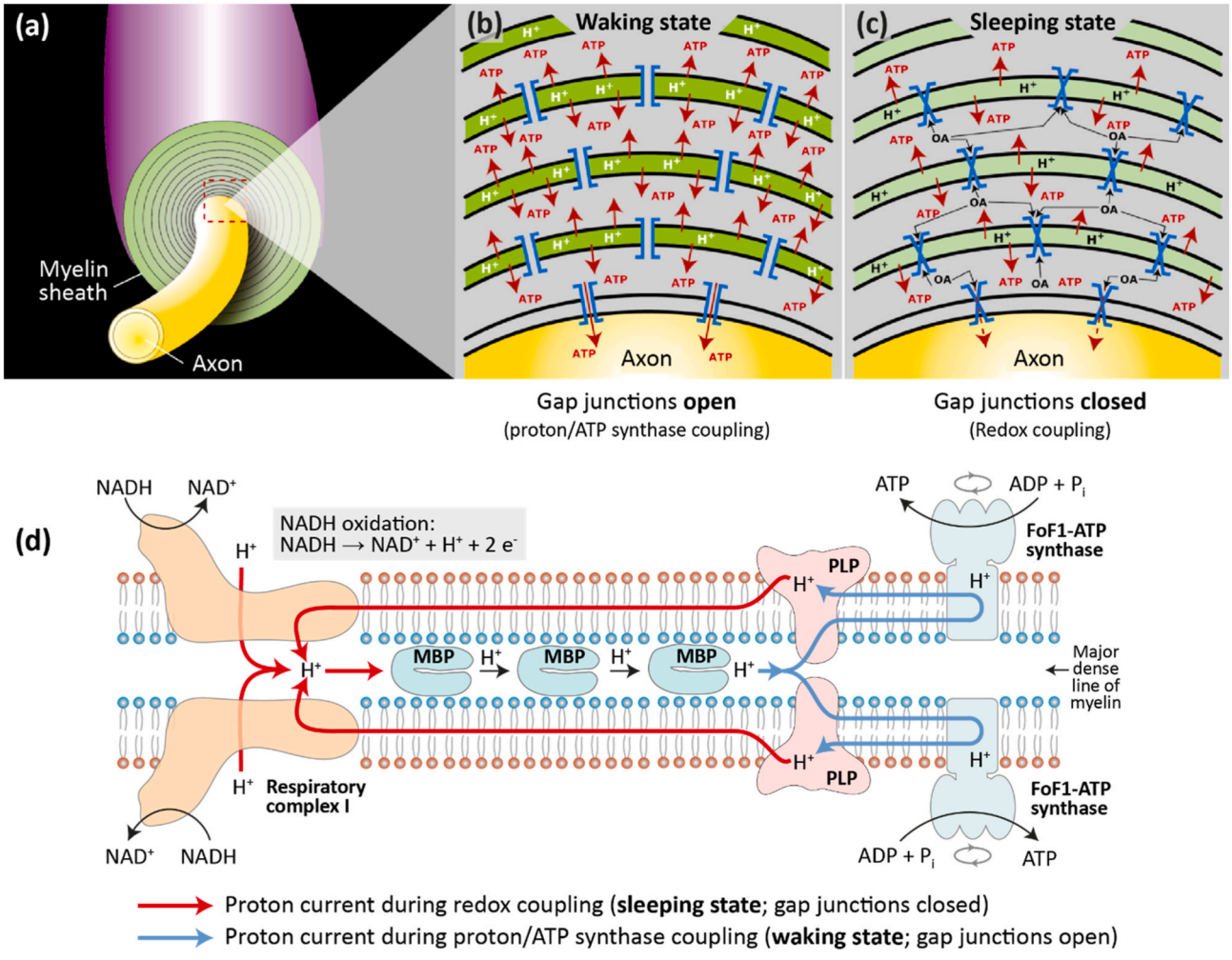


Fig. 4. The significance of gap junctions in the proposed proton current cycle in myelin. (a) Cross section of the myelin sheath around the axon. **(b)** During wakefulness, the major dense axon is saturated (charged) with protons (dark green color) and is progressively "discharged" by proton/ATP synthase coupling. The gap junctions are open. **(c)** During sleep, oleamide (OA) closes the gap junctions and the "discharged" MBP and PLP (light green color) are progressively "recharged" by the redox coupling. **(d)** Illustration of proton currents during sleep and wakefulness. The red line indicates the proton path of the redox coupling phase, which "recharges" the MBP and the major dense line base domain of the PLP. This is the typical sleep phase. The purple line indicates the proton transfer from the MBP (and the base domain of the PLP) to the intraperiod base domain of the PLP, which drives the rotation of Fo of FoF1-ATP synthase to produce ATP. This pathway is operational during wakefulness. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

In a nutshell, the MBP and the major dense line basic domain of the PLP are loaded with positive charges during sleep, while during wakefulness ATP is synthesized with the translocation of the positive charges from the MBP and the major dense line basic domain of the PLP that rotate the Fo rotor of the ATP synthase across the intramembrane proton conductor domain of the PLP and the protons accumulate in the intraperiod basic domain of the PLP and probably on the nitrogenous residues (i.e., choline moiety) of the phospholipids facing the intraperiod line. This pathway is indicated by a dotted purple line in Fig. 3(b) and a solid purple line in Fig. 4.

Thus, sleep removes protons from the intraperiod basic domain of the PLP (and probably from the polar residues of the membranes at the sides of the intraperiod line) to transfer them to the MBP, whereas wakefulness, and even more so sleep deprivation, brings them to the domain of the PLP that is in the intraperiod line. With sleep deprivation, the MBP consequently loses the positive charges on the two opposite domains which then no longer repel each other due to the positive charges typical of a myelin recharged during sleep, with a consequent decrease in myelin thickness, which has been observed (Bellesi et al., 2018; de Vivo and Bellesi, 2019). Furthermore, it has been shown that sleep increases the volume of grey matter, while sleep deprivation decreases it (Dai et al., 2018; Sun et al., 2020), in line with a decrease in the distance between the layers of myelin due to the reduced repulsion between charges of the same sign (+) following sleep deprivation.

5. Gap junctions of the myelin sheath control the influx of ATP into the axon

Myelin is rich in gap junctions, as shown by proteomic analyses (Siems et al., 2020). In the brain, myelin is the structure with the highest concentration of gap junctions (Nagy et al., 2003). Gap junction mutations are associated with severe pathologies (Belousov et al., 2018; Kleopa, 2011). In our laboratories, we have demonstrated that gap junctions control the speed of nerve impulses (Adriano et al., 2011; Ravera et al., 2016b). According to our model, the gap junctions can be in an open state, allowing the influx of ATP into the axon, while in a closed state the axon no longer receives ATP from the myelin.

It has been shown that oleamide (OA) [or cerebrodiene (Lerner et al., 1994)] is a gap junction blocker (Guan et al., 1997) and an inducer of sleep (Mendelson and Basile, 2001), supporting the relevance of gap junctions in our hypothesis about the proton current circuit in myelin and extra-mitochondrial OXPHOS. OA formation is catalyzed by cytochrome *c* (Driscoll et al., 2007).

With these premises we can formulate the hypothesis that myelin acts as a proton capacitor, accumulating energy in the form of protons during sleep to convert it into ATP during wakefulness (Fig. 4). In the sleeping state (Fig. 4(c)), the large dense fiber is poorly protonated (low intensity of green color) and is destined to “charge” because (i) the redox coupling is active involving proton transfer to the MBP and PLP located in the large dense fibers and because (ii) the formation and passage of ATP to the axon is blocked by OA, which closes the gap junctions, so that proton flow at the Fo of the ATP synthase is blocked and accumulates in the MBP and PLP. During the waking state (Fig. 4(b)), the main dense lines are saturated with protons (strong green intensity of the main dense line) and the proton/ATP synthase coupling is operational with the transfer of protons to the Fo rotor of the ATP synthase, triggering synthesis of ATP. Protons have free access to the axon thanks to the absence of OA, which would close the gap junctions.

OA plays an important role in favor of our hypothesis (Boger et al., 1998b). In our laboratory, we have demonstrated that ATP formation by myelin vesicles decreases from about 48 to about 16 nmol/min/mg protein in the presence of 400 μM OA (Ravera et al., 2016b). It has been shown that OA accumulates during wakefulness (Mendelson and Basile, 2001) and that OA is a naturally occurring hypnotic (Boger et al., 1998a) and induces sleep when injected into rats (Cravatt et al., 1995). The neuropharmacological effects of OA have been demonstrated in male and female mice (Akanmu et al., 2007). Interestingly, cytochrome *c* catalyzes the formation of OA (Boger et al., 1998a; Driscoll et al., 2007) and therefore it can be speculated that it is formed as a result of active OXPHOS, which is known to involve cytochrome, transporting electrons from respiratory complex III to respiratory complex IV. Hydrolysis of OA can occur by fatty acid amide hydrolase, which is strongly inhibited by 2-octyl-gamma-bromoacetoacetate, a compound that acts as a hypnotic (Boger et al., 2000), confirming the hypnotic effect of OA.

It is interesting to note that, surprisingly, OA has not yet found application in the pharmacological field as a sleep aid, although it is a natural sleep inducer.

6. Calculation of the proton storage capacity of myelin in relation to sleep times

After the structural foundations of the proton-capacitor hypothesis of MBP/PLP and the proton current circuit between the myelin proteins and the FoF1-ATp synthase were explained in the previous sections, the following section calculates the number of protons that can be moved in 1 cm^3 of myelin by oxidative metabolism during 8 h of sleep. This allows then a comparison with the number of protons that laminar myelin buffers can take up to saturation, as explained in section 2.

6.1. Calculation of the number of protons that can be transferred by the metabolism of 1 cm^3 of myelin in 8 h of sleep

The human brain has been found to consume an average of 47.25 cm^3 of O_2/min in the waking state (Siwicka-Gieroba et al., 2022). During sleep, O_2 consumption decreases by 21 % (Caporale et al., 2021), resulting in an O_2 consumption rate of 37.32 cm^3 O_2/min. Moreover, a further subtraction of 7 % is necessary due to the phenomenon known as the O_2 paradox, which is due to O_2 consumption not corresponding to the stoichiometry of glucose combustion. Taken together, this results in an O_2 consumption rate of 34.71 O_2/min during sleep (i.e., about 75 % of the rate during wakefulness).

The volume of 1 mol of gas is 22,414 l, therefore a human brain consumes 1.54 mmol of O_2/min during sleep. Given an average weight of the human brain of 1350 g (Hartmann et al., 1994), the consumption is 1.14 μmol O_2/g/min, which can be equated to 1 g on the assumption that 1 g is ~1 cm^3 of brain. Myelin constitutes 40 % of the brain (Morell and Norton, 1980), therefore 1 cm^3 of myelin will consume $1.14 \times 0.4 = 0.456$ μmol O_2/g/min. The result is multiplied by two to obtain micromoles of atomic O_2, with the assumption that myelin consumes O_2 in a manner analogous to the average brain. It is reasonable to assume that half of the O_2 consumed by myelin is used for basal metabolism and the other half for the charging of protons onto the proton capacitor of the myelin itself. This yields the following result: $(0.456 \times 2)/2 = 0.456$ μmol of atomic O_2/min/cm^3 of myelin. By multiplying this figure by 60 (minutes in an hour) and by 8 (hours of sleep), we obtain the result that 218.9 μmol of atomic oxygen are consumed by 1 cm^3 of myelin in 8 h of sleep.

The consumption of each O_2 atom involves the movement of ~12 protons (Deshpande and Mohiuddin, 2024). Consequently, the protons undergo a movement of $218.9 \times 12 = 2627$ μmol in 8 h of sleep by 1 cm^3 of myelin. Upon multiplying this value by 10^{-6} to obtain moles and by Avogadro’s number (6.06×10^{23}), it can be determined that ~15.9×10^{20} protons are moved in 1 cm^3 of myelin over the course of 8 h of sleep, rounded to ~16×10^{20}.

6.2. Calculation of protons that can saturate one cm^3 volume of myelin

Each molecule of MBP has the capacity to absorb protons through the charged residues (lysine, arginine and histidine), with a total of 61 residues per MBP molecule. Furthermore, PLP (the most abundant protein in CNS myelin), which binds to the MBP, may also function as a buffer. It contains 18 basic residues (see Fig. 3(a)), which must be

multiplied by two because two of these domains overlook the major dense line, resulting in 36 + 61 = 96 of the MBP. It is then necessary to consider the buffering properties of the peptide bond, which are present in 304 instances in the MBP and 65 times two, or 130 instances, in the PLP. Therefore, the total number of absorbable protons per MBP molecule is 531, comprising 97 from the MBP itself and 304 from the peptide bonds. Additionally, the heads of the phospholipids, which exhibit a surface area on the membrane that is undoubtedly greater than that of MBP, can also be protonated. This suggests that the capacity of an MBP molecule may be ~600 protons.

Next, we turn to calculating the surface area of the major dense line in 1 cm^3 of myelin.

Given the MBL length of about 5 nm, it can be approximated that MBP occupies an area of 5 nm × 5 nm = 25 nm^2 in the major dense line of myelin. The repetitive distance has been reported to be 16 nm by electron microscopy (Morell and Quarles, 1999). In 1 cm^3 of myelin, the surface area of the major dense line will be 1 cm^2, repeated for 1 cm in height as many times as is required to achieve the ratio 0.01 m divided by the repetitive distance 16 nm. Therefore, 1×10^{-2} m divided by 16×10^{-9} m provides 6.25×10^5 layers of myelin. Consequently, the cm^2 must be multiplied by 6.25×10^5. The total surface area of the major dense line in 1 cm^3 of myelin is therefore 6.25×10^5 cm^2. To convert this to m^2, the figure has to be multiplied by 10^{-4}. The total surface area of the major dense line in 1 cm^3 of myelin is ~62 m^2.

By dividing the surface area of the major dense line in 1 cm^3 of myelin by the surface area occupied by one molecule of MBP (which, as an approximation, can occupy half of it), it is possible to calculate the number of MBP molecules that can be embedded: $(62/2)/25 \times 10^{-18} = 1.24 \times 10^{18}$. To get the final result, this value must be multiplied by 600, which gives 744×10^{18}, i.e., $\sim 8 \times 10^{20}$. Moreover, it is essential to consider the potential protonation of phospholipids on the two cytosolic sides of the major dense line. A phospholipid occupies an area of 0.62 nm^2 (Nagle and Tristram-Nagle, 2000), therefore, in 62 m^2 of the major dense line, they will occupy 62 m^2: $0.62 \times 10^{-18} = 1 \times 10^{20}$, which must be multiplied by 2 because the major dense line is delimited from 2 membranes, i.e., 2×10^{20}. Therefore, the total number is $\sim 8 \times 10^{20} + 2 \times 10^{20} = \sim 10 \times 10^{20}$, which is of a similar order of magnitude to the $\sim 16 \times 10^{20}$ protons transferred by catabolism in 1 cm^3 of myelin obtained in the previous section.

In order to formulate an overall proton balance, it is necessary to consider the charge balance associated with the synthesis of ATP: P_i (3 negative charges) + ADP (3 negative charges) → ATP (4 negative charges).

This balance of negative charges, i.e., 3 + 3–4, produces 2 negative charges that disappear, which can be equated to 2 positive charges that are formed with the synthesis of ATP in the intraperiod line. These positive charges may contribute to the flow of positive charges supported by the respiratory complexes in the redox coupling phase during the sleeping state.

The calculation presented here should be seen as an initial approximation; further, more detailed calculations are necessary. Nevertheless, the calculations set out here support the hypothesis presented in this article about the proton current circuit between the myelin proteins and FoF1-ATP synthase that possibly is present in myelin.

7. The duration of sleep depends on the amount of white matter and neuronal energy consumption

We investigated whether there is a correlation between the white matter volume (WMV) in the brain and the sleep duration (% of 24 h) in mammals. We were able to find and analyze data from 23 mammals. Indeed, there is an allometric scaling between WMV and sleep time ($r = 0.5562$, $p < 0.01$; Fig. 5).

Given that the white matter is composed of over 95 % myelin, at least in humans (Huitema et al., 2021), it can be postulated that this correlation is also true for myelin in the brain and sleep time.

WMV (i.e., myelin) is therefore negatively correlated with sleep time. This can be interpreted as an indication that a lack of myelin results in a reduction in the energy required for waking activities. An increased sleep time (% of 24 h) of a mammal might therefore be a consequence of a reduced energy contribution from myelin due to its specific WMV.

This allometric scaling also seems to apply to humans over the lifespan: the WMV follows an inverted U-shaped function over the lifespan (Sowell et al., 2003), and the sleep duration (% of 24 h) follows a

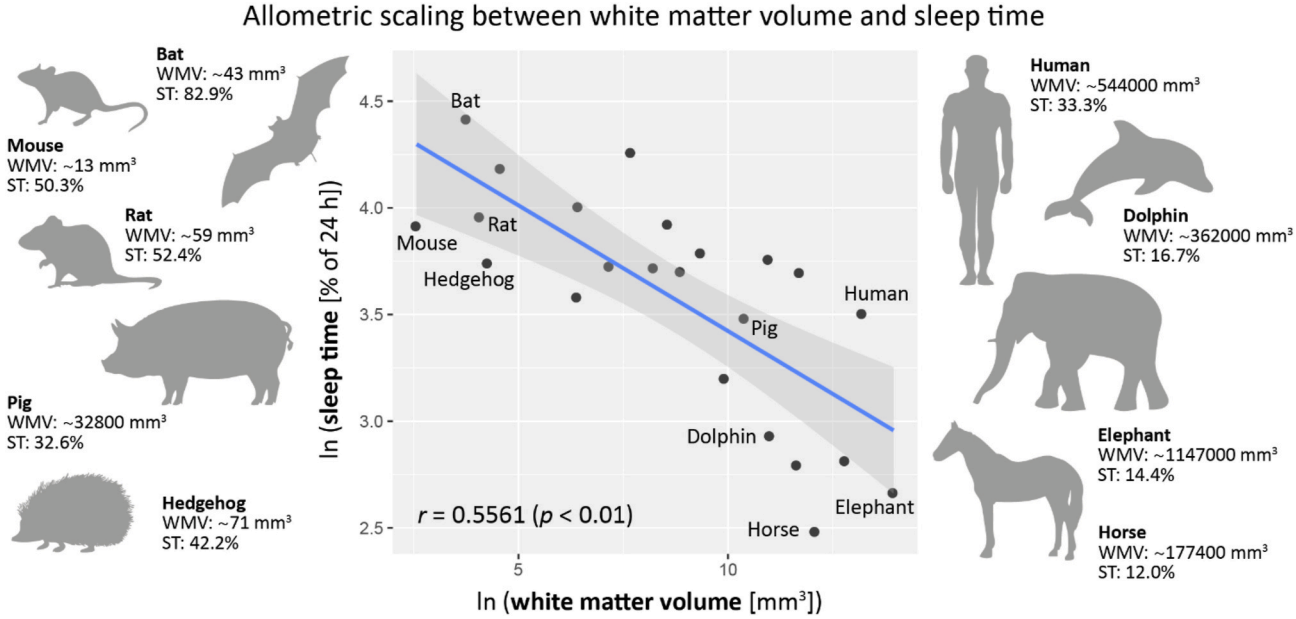


Fig. 5. Allometric scaling between white matter volume and sleep time. Data for 23 mammals. Data for sleep time obtained from tabulated values by Eric H. Chudler (University of Washington, USA) based on published data (Aserinsky, 1999; Campbell and Tobler, 1984; Gravett et al., 2017; Kryger et al., 1989; Siegel, 2022; Tobler, 1989). Sleep time data for the lesser mouse lemur taken from Hozer et al. (Hozer et al., 2019). White matter volume data obtained from Thang and Sejnowski (Zhang and Sejnowski, 2000) and Mota et al. (Mota et al., 2019). Data from 9 mammals are highlighted in the plot and by visualization as examples for mammals from both ends of the allometric scaling law. WMV: white matter volume, ST: sleep time.

U-shaped function (Coutrot et al., 2022). Between the ages of about 30–50, WMV reaches its highest values, while sleep time (% of 24 h) reaches minimum in this period.

Interestingly, the relationship between WMV and sleep time is also evident when comparing male and female humans: women have less white matter than men (Gur et al., 1999) and require more hours of sleep on average (Burgard and Ailshire, 2013).

8. Summary, conclusions and outlook

The theory presented here about the underlying bioenergetic function of sleep is consistent with the widely accepted notion that sleep is a process that builds up energy reserves. The above description emphasizes the centrality of mitochondrial components (but not necessarily whole mitochondria) in cell energetics exacerbated by the relocation of the typical OXPHOS machinery to extra-mitochondrial sites, and myelin is an excellent example of this. This process presumably occurs with the formation of MDV fusion or association with myelin. This is certainly a possibility since we have previously shown that MDV can fuse and complement ATPase-deficient mitochondria (Hazan Ben-Menachem et al., 2023). It is hypothesized that the metabolism of glucose, the primary nutrient for the brain, accumulates energy within the myelin sheath during periods of sleep and releases it during periods of wakefulness. This process makes use of the glycolytic catabolism recently summarized by us (Morelli and Scholkmann, 2024), which involves the pentose phosphate pathway (PPP) as the main metabolic pathway of glucose, with the possibility of recycling (Rodriguez-Rodriguez et al., 2013). The modified glycolysis pathway is therefore a functional explanation of carbohydrates' lipogenic character. It is noteworthy that lipid synthesis by myelin has also been demonstrated (Asadollahi et al., 2022). This process is increased, and lipids serve as an excellent energy reserve for the body. It can thus be concluded that myelin plays a dual role in the accumulation of energy during sleep, acting as a physical reservoir for energy in the proton capacitor and a chemical reservoir for energy in form of lipids that are particularly abundant in myelin. Indeed, it has been demonstrated that intense physical and neural activity, such as that experienced by participants in the New York Marathon, results in a notable reduction in myelin (Ramos-Cabrer et al., 2023). Sleep deprivation also leads to myelin degradation (Bellesi et al., 2018).

The present study investigated the potential for the two most abundant myelin proteins, PLP and MBP, to form a coupled system that could facilitate energy storage by protons during sleep, thereby providing a reserve for the brain during wakefulness. Additionally, the study aimed to elucidate the integration of this process with the mitochondrial molecular machinery, which is known to be fully functional and present in myelin (Ravera et al., 2021). The mitochondrial properties of the myelin sheath were considered to be those of a concentric multilamellar structure, capable of carrying out OXPHOS with associated O_2 consumption and ATP production. It can therefore be surmised that mitochondria are involved in sleep, albeit in a modified form, as has been previously reported by several research groups (Aboufares El Alaoui et al., 2023; Richardson and Mailloux, 2023; Sarnataro et al., 2024). However, these are likely to be extra-mitochondrial systems that originate from the mitochondria themselves. Additionally, myelin is capable of readily absorbing O_2 due to its high lipid content (Morelli and Scholkmann, 2023). Moreover, the presence of gap junctions in the myelin sheath enables the transfer of ATP to the axon, thereby facilitating nerve conduction during wakefulness. Conversely, during sleep, these gap junctions are closed by OA, which accumulates during wakefulness.

In developing this model, consideration was given to the substantial abundance of MBP in the major dense line. Given its basic nature, MBP can receive and accumulate protons, thereby functioning as a proton capacitor. Moreover, PLP possesses domains that are capable of functioning as proton capacitors. Additionally, PLP contains a domain comprising the amino acids threonine and serine, which are highly effective proton conductors. This confers upon PLP the function of a proton translocator between the major dense line and the intraperiod line, keeping in mind that protons are never free but rather coupled to structures that absorb them, such as the basic domains of both MBP and PLP. So the hypothesis is based on the premise of direct proton transfer between membrane-immersed buffers, a phenomenon that is challenging to verify experimentally. This particular hypothesis has been also proposed by Georgievskii et al., (2002). Furthermore, the presence of proton buffers has been demonstrated in thylakoid membranes, which are "*sequestered domains in which protons are held in a metastable state out of equilibrium with those in the inner (lumen) or the outer aqueous bulk phase*" (Theg et al., 1988). The existence of non-bilayer structures has also been demonstrated in internal mitochondrial membranes (Garab et al., 2022; Gasanoff et al., 2021; Li and Gasanoff, 2023). These structures have been shown to perform the function of a proton buffer, which lends support to the hypothesis that "proton buffers" are a general feature of all cells. The quantity of these structures should depend on the amount of energy processed, and myelin certainly responds to this criterion, given that it is home to the greatest quantity of energy from part of the organism.

It is interesting to note that an increase in ATP synthesized by FoF1-ATP synthase is associated with an increase in non-bilayer structures, which is experimentally linked to a possible increase in the specific proton capacitor of non-bilayer structures (Garab et al., 2022). This association between non-bilayer structures and proton capacitor function has been verified by studying the effect of toxins from the spider *Latrodectus pallidus* venom, which are homologous to the MBP structure (Chaudary et al., 2024). The link between non-bilayer structures and the proton-capacitor function of myelin is an interesting topic that warrants further attention.

It is therefore possible to propose an ATP synthesis with a proton current circuit comprising two distinct phases. The first is a "redox coupling" phase (Morelli et al. 2019), during which the respiratory complexes, in conjunction with oxygen consumption, can accumulate protons within the MBP and the major dense line basic domain of the PLP. The second is a "proton/ATP synthase coupling" phase, during which the accumulated protons are discharged onto the ATP synthase, resulting in the subsequent synthesis of ATP. A pivotal function is possibly performed by OA, which has been demonstrated to be a natural hypnotic (Boger et al., 1998a). OA accumulates during wakefulness and closes the gap junctions, preventing ATP from reaching the axon and thereby determining the quiescence of nervous activity, which is a defining characteristic of sleep.

According to our hypothesis, as sleep progresses, MBP and PLP are progressively loaded with protons until saturation. The order of the size of the protons that can saturate the laminar buffers containing MBP and PLP in 1 cm^3 of myelin is the same as that observed during sleep. It is notable that OA should be eliminated through conversion to the innocuous oleic acid, given that it undergoes spontaneous deamidation. This process has been demonstrated to function as a molecular biological clock (Robinson and Robinson, 2004), which lends plausibility to this hypothesis. Deamidation is a general process in biology, to the extent that it has been proposed as a molecular clock and a potential precursor to protein removal (Panfoli et al., 2010).

The waking phase is characterized by the opening of the gap junction, which allows the ATP synthase to receive a flow of protons released by the laminar buffers, which are composed of MBP and the major dense line basic domain of the PLP. This results in a "proton/ATP synthase coupling" with a semi-hypothesized circuit that begins at the laminar layers of the major dense line and passes through the Fo rotor, before reaching the intramembrane proton conductor domain of the PLP. This has been demonstrated to be an efficient proton conductor (Lin and Lees, 1982), accumulating in the intraperiod basic domain of the PLP, which faces the intraperiod line. Upon the discharge of the laminar buffer of protons, the synthesis of reserve ATP is no longer possible, resulting in the cessation of nerve conduction and the onset of the sleep phase.

This model of the "proton capacitor" responds to the necessity for the components that accumulate energy to be present in sufficient quantities, which is a defining characteristic of all accumulators. Indeed, the principal proteins that serve as proton capacitors, namely MBP and PLP, are the most prevalent proteins in myelin. The distribution of basic amino acids that can accept protons is situated in optimal locations to receive and transfer protons. The model is also supported by OA, whose hypnotic action is well documented and specifically aimed at blocking gap junctions. It is important to note that the molecular hypnotic action of OA in our model can now be explained satisfactorily.

The recent emphasis on the role of mitochondria in sleep regulation (C. Hartmann and Kempf, 2023) is supported by our model of the myelin proton capacitor. The model places significant emphasis on mitochondrial function, rather than on the mitochondria themselves within nervous tissues. This is because mitochondrial function is carried out by extra-mitochondrial structures, with myelin playing a fundamental role as an "energy administrator" and therefore being pivotal in the formulation of the sleep model described here.

In light of the aforementioned considerations, it is also possible to provide a response to the unexpected data published recently by Haynes et al., (2024). The findings of this study demonstrated that Drosophila lacking sleep for a single night exhibited elevated levels of oxidation in glia, but not in neuronal mitochondria. Even treatment with rotenone, an inhibitor of OXPHOS, yielded results exclusively in glia, with no discernible impact on neurons. Additionally, a recent report (Sarnataro et al., 2024) posits that the mitochondria do not play a central role in sleep regulation. This is based on the observation that the induction or prevention of mitochondrial fission or fusion alters sleep in Drosophila, and that myelin contains the mitochondrial components OXPHOS. In a similar vein, it has been observed that treatment with the plant decoupler dinitrophenol in mice induces the same symptoms as sleep deprivation (Kalinchuk et al., 2003). Furthermore, sleep deprivation has been shown to increase the activity of certain respiratory complexes (Nikonova et al., 2010).

In conclusion, the presented model allows for the examination of myelin as a glial component that functions as a proton capacitor, aligning with the widely accepted notion that sleep serves to recharge the nervous system with energy.

CRediT authorship contribution statement

Alessandro Maria Morelli: Writing – original draft, Writing – review & editing, Validation, Conceptualization. **Ann Saada:** Writing – review & editing, Validation. **Felix Scholkmann:** Writing – review & editing, Visualization, Validation, Formal analysis, Conceptualization.

Data availability

No datasets were generated during the current study.

Ethics approval

Not applicable.

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Declaration of competing interest

The authors declare that there are no competing interests associated with the manuscript.

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